

OPERATING SYSTEMS

Storage Management

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OPERATING SYSTEMS

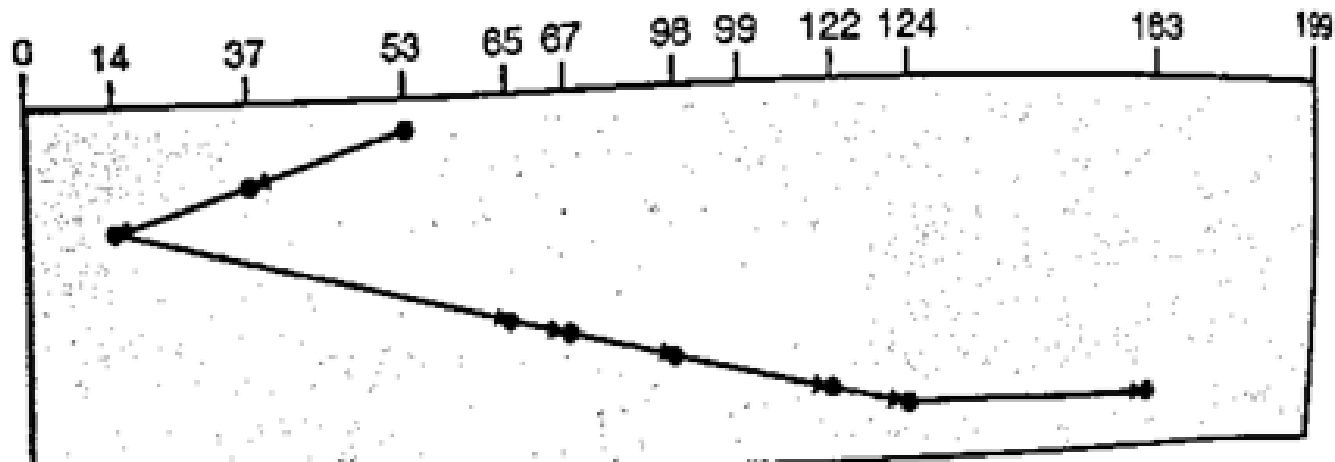
Disk Scheduling Algorithm - LOOK

LOOK scheduling improves upon SCAN by looking ahead at the queue of pending requests, and not moving the heads any farther towards the end of the disk than is necessary.



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Disk Scheduling Algorithm - LOOK



LOOK Scheduling

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Disk Scheduling Algorithm - LOOK

$$\begin{aligned}\text{Total Head movement} &= |53-37| + |37-14| + |14-65| \\ &\quad |65-67| + |67-98| + |98-122| + \\ &\quad |122-124| + |124-183| \\ &= 16 + 23 + 51 + 2 + 31 + 24 + 2 + 59 \\ &= 208\end{aligned}$$

Head movement shows 208 cylinders

Advantages

- Throughput increases as it is not moving to the end of the disk

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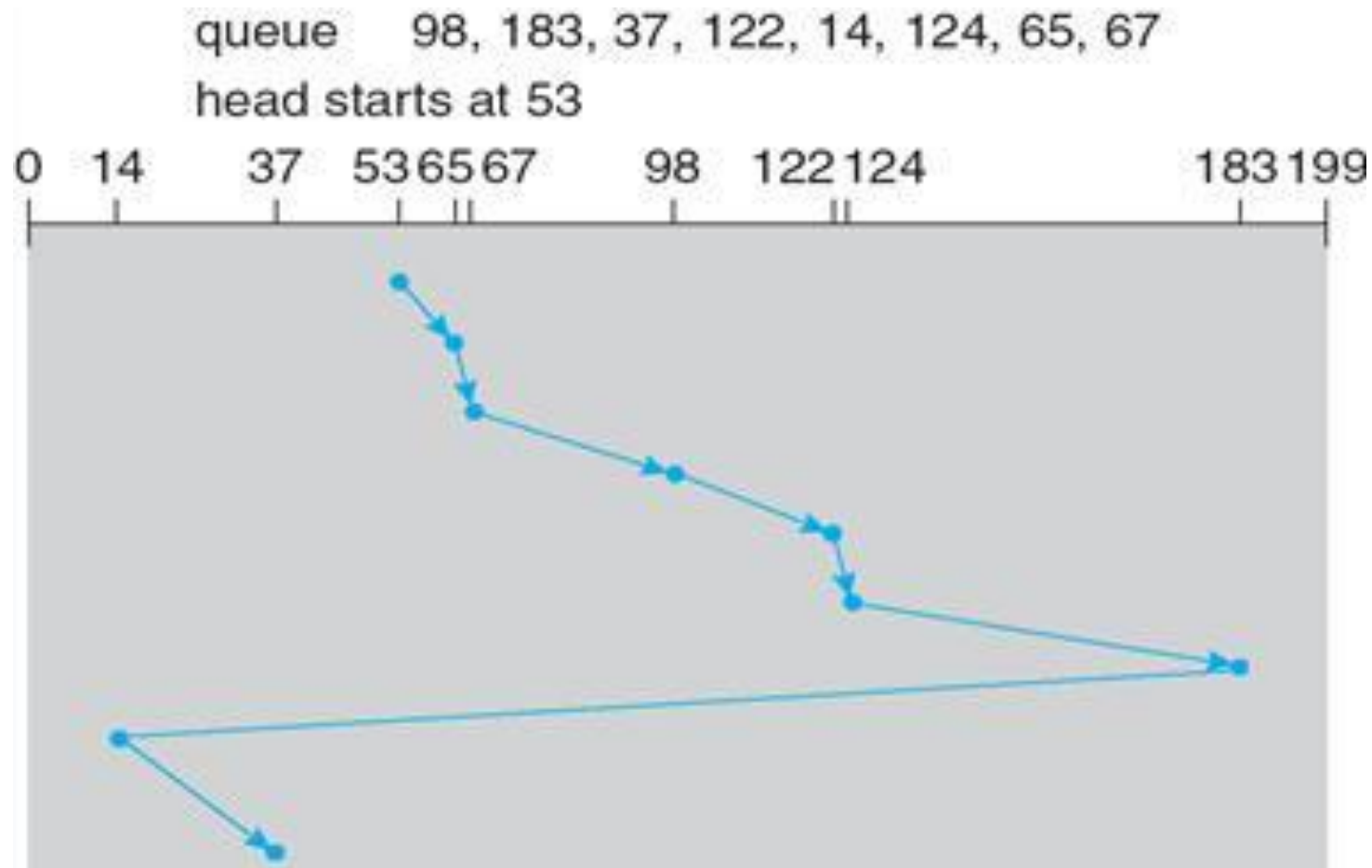
Disk Scheduling Algorithm - CLOOK

Version of C-SCAN

Arm only goes as far as the **last request** in each direction, then reverses direction immediately, without first going all the way to the end of the disk.

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Disk Scheduling Algorithm - CLOOK



Head movement shows 322 cylinders

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Disk Scheduling Algorithm – C-LOOK



$$\begin{aligned}\text{Total Head movement} = & |53-65| + |65-67| + \\ & |67-98| + |98-122| + \\ & |122-124| + |124-183| + \\ & |183-14| + |14-37|\end{aligned}$$

$$\begin{aligned}&= 12 + 2 + 31 + 24 + 2 + 59 + 169 + 23 \\ &= 322\end{aligned}$$

Head movement shows 322 cylinders

Advantages

- Maximizes locality and resource utilization

Disadvantages

- Can seem a little unfair to other requests and if new requests keep coming in, it cause starvation to the old and existing ones.

Selecting a Disk Scheduling Algorithm

SSTF is common and has a natural appeal

SCAN and **C-SCAN** perform better for systems that place a heavy load on the disk

Either SSTF or LOOK is a reasonable choice for default algorithm



THANK YOU

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